Unit 4 Review
AP Calculus Name_____

YOU MUST SHOW ALL WORK TO RECEIVE CREDIT.

1. (Calculator Active) A bottling company produces bottles of water. The amount of bottles that have been produced at time t hours can be modeled by $B(t) = 3125t^{1.5} + 2t$ where $0 \le t \le 10$.

(a) Find $B'(4)$. Using correct units, interpret the meaning of this value in context of the problem.

(b) What are the units of $B''(3)$? Do not find $B''(3)$.

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2. (Calculator Active) For $0 \le t \le 8$ hours, the number of people in line to vote at a local precinct is defined by $L(t) = 78\sin\left(\frac{t}{13}\right) + 15$. Find $L'(5)$. Using correct units, interpret the meaning of this value in context of the problem.

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t (minutes)	1	3	6	8
$D(t)$ (donuts/minute)	21	46	35	12

3. For $0 \le t \le 8$ minutes, the rate that a local bakery sells donuts, in donuts per minute, can be modeled by the differentiable function $D(t)$. Selected values for $D(t)$ are shown in the table above.

(a) Approximate $D'(2)$. Using correct units, interpret the meaning of $D'(2)$ in context of the problem.

(b) Is there a time c, such that $1 < c < 8$, when $D(c) = 15$? Justify your answer.

(c) It is known that the bakery has sold 220 donuts at time $t = 8$. Using a linear approximation at $t = 8$, estimate the number of donuts the bakery has sold at time $t = 10$ minutes.

t (hours)	0	3	4	6
$P(t)$ (100's people/hour)	18	51	46	82

4. During the time leading up to the 2024 Presidential election, people made posts on social media showing that they voted. In one town, the rate that people posted on social media, in 100's of people per hour, is modeled by $P(t)$, where t is measured in hours and P is measured in 100's of people per hour. Estimate $P'(5)$. Using correct units, interpret the meaning of this value in the context of the problem.

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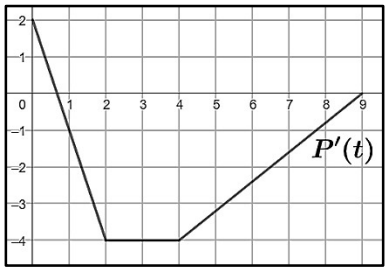
5. After a ride ends, the rate that the number of people on a carnival ride is changing is modeled by the function $P'(t)$, where $P'(t)$ is measured in people per second and t is measured in seconds. The graph of P' consists of three line segments as shown at the right.



(a) Find $P'(1)$. Using correct units, interpret the meaning of this value in the context of the problem.

(b) Find $P''(5)$. Using correct units, interpret the meaning of this value in the context of the problem.

(c) The total weight of people on the ride, in pounds, at time t seconds, is modeled by $W(t) = 4150 + 150P(t)$. Find $W'(3)$. Using correct units, explain the meaning of this value in the context of the problem.

6. (Calculator Active) For $0 \leq t \leq 10$ days, the rate that a company sells t − shirts, in 100′s shirts per day, on its website after an appearance on Shark Tank is modeled by $S(t) = 124e^{-2t}$. At what rate is the rate of t − shirt sales on the website changing at $t = 5$ days?

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7. The temperature of a cooler T, in °F, is a function of the amount of ice in the cooler i, measured in pounds, and can be written as $T(i)$. What are units for $T'(i)$ and $T''(i)$?

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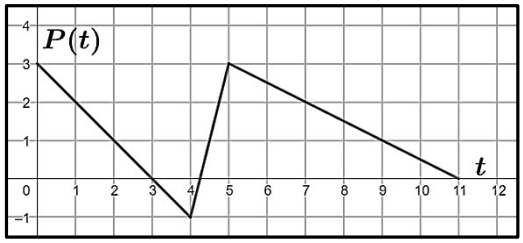
8. Let P be the rate that the number of people in a store is changing, where P is measured in people per second and t is measured in seconds. The graph of P is shown at the right. Find $P'(2)$. Include units of measure.



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9. For $t \geq 0$, a particle moves along the x axis with velocity $v(t) = t^2 - 7t + 10$ and $x(6) = -4$.

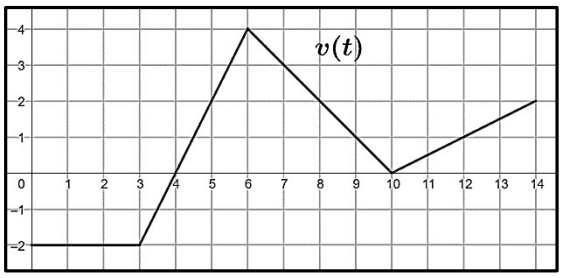
(a) Find all times t when the particle changes direction. Give a reason for your answer.

(b) Find the velocity of the particle at the time when its acceleration is 1.

(c) Find $a(3)$. Is the particle speeding up or slowing down when $t = 3$? Give a reason for your answer.

(d) Is the particle moving towards or away from the origin when $t = 6$? Give a reason for your answer.

10. For $0 \le t \le 14$ seconds, a particle moves along a horizontal axis with velocity modeled by $v(t)$, measured in meters per second. The graph of v is shown at the right and consists of four line segments.



(a) On what interval(s) is the particle moving left? Explain your reasoning.

(b) Is the particle speeding up, slowing down, or neither when $t = 3.5$? Give a reason for your answer.

(c) Find the average acceleration of the particle over the interval $0 \le t \le 14$ seconds. Include units of measure.

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11. (Calculator Active) For $0 \le t \le 2$, a bug moves along a straight path with velocity $v(t) = 2^t \cos(5 \ln(t + 1))$.

(a) At what time(s) t does the bug turn around? Give a reason for your answer.

(b) On what interval(s) does the bug have a negative acceleration?

(c) Find the velocity of the bug when $a(t) = 3$.

(d) On what intervals is the bug speeding up? Give a reason for your answer.

t (seconds)	0	3	5	7
$v(t)$ (in/sec)	-5	4	0	-12

12. For $t \geq 0$, a particle moves along the x axis. Selected values of the velocity of the particle, in inches per second, are shown in the table above where v is a differentiable function.

(a) Using the data in the table, approximate the particle's acceleration at $t = 4$ seconds. Include units of measure.

(b) Must there be a time c, where $0 < c < 9$ such that $v(c) = 6$? Give a reason for your answer.

(c) For $5 \leq t \leq 10$, the acceleration of the particle is known to be $a(t) = k - 2t$ where k is a constant. Find all possible values of k such that the particle is speeding up when $t = 7$.

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13. The area of a circle is decreasing at a rate of 3 inches squared per second. Find the rate that the radius of the circle is changing when the area of the circle is 16π inches squared.

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14. The radius of a circle changes at a constant rate of 5 meters per hour. Find the rate that the area of the circle is changing when the circumference of the circle is 18π meters.

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15. Let $2z = (4w - 2)^3$. If $\frac{dw}{dt} = \frac{1}{2}$, find $\frac{dz}{dt}$ when $z = 4$.

16. Katelynn is chewing bubble gum and begins to blow a bubble that is spherical. The size of the bubble increases at a rate of 2 cm^3 per second. At what rate is the radius of the bubble increasing when the volume of the bubble is $\frac{32\pi}{3}$ cm^3. $\left(V = \frac{4}{3}\pi r^3\right)$

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17. The height of a child, in inches, can be modeled by the function $h(w) = 2\sqrt{4w - 119} + 30$, where w is the weight of the child, in pounds. When the child is 48 inches tall, their weight is increasing at a rate of 8 pounds per year. What is the rate of change of the height of the child with respect to time, in inches per year, at the time when the child is 48 inches tall?

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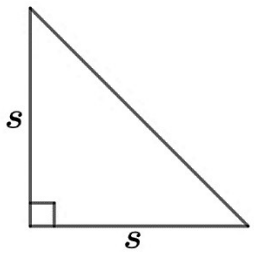
18. A plastic pond is being filled with water when a leak forms at the bottom of the pond, causing water to drain out. The volume of water, in in^3, in the pool can be modeled by the function $V(h) = \frac{h^3}{15}$ where h is the height of water inside the pond in inches. The rate that the height of the water is changing is given by $\frac{dh}{dt} = -2\sqrt{h}$, where h is measured in inches and t is measured in seconds. Find the rate that the volume of water in the pond is changing with respect to time, when the height of the water is 4 inches.

19. The area of an isosceles right triangle with legs of length s is decreasing at a rate of 6 cm^2 per second. Find the rate that the length of the leg is changing when the area of the triangle is 32 cm^2.



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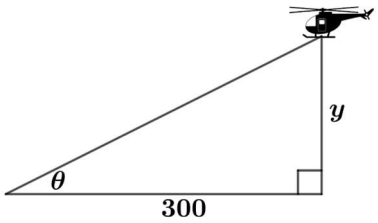
20. A helicopter lifts off the ground vertically at a contant rate of 8 feet per second. An observer watches the helicopter lift off the ground at a point 300 feet away from the ground directly below the helicopter.



(a) Find the rate that the distance between the observer and the helicopter is changing, in feet per second, when the helicopter is 125 feet above the ground.

(b) Let θ be the angle of elevation from the observer to the helicopter as shown in the figure above. Find the rate of change of θ when $y = 400$ feet.

(c) Once the helicopter reaches a height of 800 feet above the ground, it begins to move horizontally away the observer at a rate of 100 feet per second. Find the rate of change, in feet per second, in the distance between the observer and the helicopter at the time when the helicopter has been flying horizontally for 12 seconds.

21. The function g is continuous and differentiable with $g(3) = -4$ and $g'(3) = 2$. Use the linear approximation for g to estimate $g(2.8)$.

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22. The function k is differentiable and it is known that $k'(x) = 2k(x) + 1$. Write an equation of the line tangent to k through $k(2) = 3$. Use the tangent line to approximate $k(2.1)$.

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x	1	3	6	7
$f(x)$	6	2	-1	-3
$f'(x)$	-2	-5	9	5
$f''(x)$	8	0	-6	4

For $23 - 26$: The function f is twice differentiable. Selected values of f, f', and f'' are given in the table above.

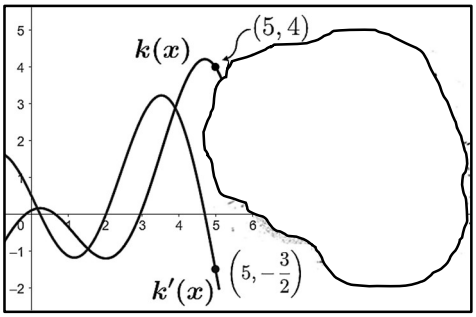
23. Write an equation of the line tangent to f at $x = 7$. Use this equation to approximate $f(7.2)$.

24. Write an equation of the line tangent to f' at $x = 1$. Use this equation to approximate $f'(0.9)$.

25. Let $g(x) = f(x^2 - x)$. Estimate $g(3.1)$ using the tangent line approximation for g at $x = 3$.

26. Let $L(x)$ be the linear approximation to f at $x = 3$. Find the value c such that $L(c) = -8$.

27. Uh oh! Mr. Carlson was trying to graph a function k and its derivative k'. However, somehow his ink pen exploded and left a giant ink stain on the graph and he does not have the equation for k or k' explicitly.



(a) Write an equation of the tangent line to k at $x = 5$ and use it to estimate $k(7)$.

(b) Based on the graphs above, do you think the estimate in part (a) is an overestimate or underestimate of $k(7)$? Give a reason for your answer.

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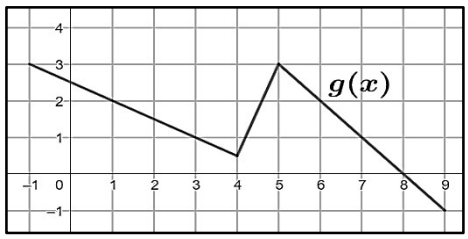
x	1	3	6	7
$f(x)$	4	2	12	0
$f'(x)$	0	-3	2	-5
$f''(x)$	3	5	-3	8



The function f is twice differentiable with selected values of f, f', and f'' shown in the table above. The graph of the function g is given above and consists of three line segments.

28. $\lim\limits_{x \to 3} \dfrac{f(x) - x + 1}{x^2 - 9}$

29. $\lim\limits_{x \to 3} \dfrac{g(2x) - 2}{f'(7 - 2x)}$

30. $\lim\limits_{x \to 7} \dfrac{f(g(x)) - 4}{\ln(2x - 13)}$

31. $\lim\limits_{x \to 6} \dfrac{f(x) - 2x}{x^2 - 12x + 36}$

32. $\lim_{x \to -3} \frac{x^2 + 3x}{e^{2x+6} - 1}$

33. $\lim_{x \to 2} \frac{\cos(x-2)}{4 - 2x}$

34. $\lim_{x \to \infty} \frac{2^x}{x^{75}}$

35. $\lim_{x \to -\infty} \frac{\sqrt{4x^2 + 5}}{3x - 1}$

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36. Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = \frac{x+2}{y^2}$ thru the point $f(0) = -1$ and $f(x)$ is twice differentiable.

a) Find $\lim_{x \to 0} \frac{[f(x)]^2 - \cos(x)}{\sin(3x)}$

b) It is known that $\lim_{x \to \infty} f(x) = \infty$. Find $\lim_{x \to \infty} \frac{[f(x)]^3}{5x^2 - 1}$.